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Rechargeable battery, electrode assembly and method for manufacturing the same

Abstract

A rechargeable battery, an electrode assembly, and an electrode assembly manufacturing method are disclosed. The electrode assembly includes: a separator configured to include a first insertion portion formed in a first direction and a second insertion portion formed in a second direction, which are alternatively stacked; a first electrode inserted into the first insertion portion; a second electrode inserted into the second insertion portion while the separator is disposed between the first electrode and the second electrode; a sealing portion configured to bond opposite surfaces of the separator into which the first electrode and the second electrode are inserted with the separator interposed therebetween; and a lead tab configured to include a first current collecting tab connected to the first electrode and a second current collecting tab connected to the second electrode.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation application of U.S. patent application Ser. No. 16/345,206, filed Apr. 25, 2019, which is a National Phase Patent Application of International Patent Application Number PCT/KR2017/013088, filed on Nov. 17, 2017, which claims priority of Korean Patent Application No. 10-2016-0153900, filed Nov. 18, 2016. The entire contents of all of which are incorporated herein by reference.

TECHNICAL FIELD
(1) The present invention relates to a rechargeable battery, an electrode assembly, and a manufacturing method of the electrode assembly, which have no risk of short-circuiting and are improved in stability.
BACKGROUND ART
(2) A rechargeable battery differs from a primary battery in that it can be repeatedly charged and discharged, while the latter is incapable of being recharged. Low-capacity rechargeable batteries are used in portable electronic devices such as mobile phones, laptop computers and camcorders, and large capacity batteries are widely used as power sources for driving a motor such as for hybrid vehicles.
(3) The rechargeable battery includes a nickel cadmium (NiCd) battery, a nickel hydrogen (NiMH) battery, a lithium (Li) battery, and a lithium ion (Li ion) rechargeable battery. In particular, the lithium ion rechargeable batteries are about three times higher in operating voltage than the nickel cadmium batteries or nickel hydride batteries, which are widely used as power sources for portable electronic equipment. Further, it is widely used because of its high energy density per unit weight.
(4) The rechargeable battery mainly uses a lithium-based oxide as a positive active material and a carbonaceous material as a negative active material. The rechargeable batteries are classified into liquid electrolyte batteries and polymer electrolyte batteries depending on a type of electrolyte, and a battery using a liquid electrolyte is referred to as a lithium ion battery, while a battery using a polymer electrolyte is referred to as a lithium polymer battery.
(5) In the case where the electrode assembly is implemented as a stacked type, the rechargeable battery may be deteriorated in stability due to a short circuit caused by changes in positions where a positive electrode and a negative electrode are stacked when a vibration is transferred thereto by external impact.
DISCLOSURE
Technical Problem
(6) An exemplary embodiment of the present invention has been made in an effort to provide a rechargeable battery, an electrode assembly, and an electrode assembly manufacturing method in which a short circuit due to an external impact does not occur.
Technical Solution
(7) An exemplary embodiment of the present invention includes: a separator configured to include a first insertion portion formed in a first direction and a second insertion portion formed in a second direction, which are alternatively stacked; a first electrode inserted into the first insertion portion; a second electrode inserted into the second insertion portion while the separator is disposed between

the first electrode and the second electrode; a sealing portion configured to bond opposite surfaces of the separator into which the first electrode and the second electrode are inserted with the separator interposed therebetween; and a lead tab configured to include a first current collecting tab connected to the first electrode and a second current collecting tab connected to the second electrode.
(8) The separator may be formed as a continuous zigzag type between the first electrode and the second electrode, the separator in which the first insertion portion and the second insertion portion are alternatively formed.
(9) The sealing portion may bond opposite surfaces of the separator with the first electrode interposed therebetween at the first insertion portion.
(10) The sealing portion may bond opposite surfaces of the separator with the second electrode interposed therebetween at the second insertion portion.
(11) The sealing portion may bond edges of the separator in which the first electrode and the second electrode are stacked in plural.
(12) The sealing portion may bond each edge of the separator of an uppermost side and a lowermost side on which the first electrode, the second electrode and the separator are stacked.
(13) The sealing portions may bond the separator by using at least one of thermal welding, ultrasonic wave welding, laser bonding, and an adhesive.
(14) An exemplary embodiment of the present invention includes: a case configured to accommodate the electrode assembly; a first electrode lead connected to the first current collecting tab; and a second electrode lead connected to the second current collecting tab.
(15) An exemplary embodiment of the present invention includes: (a) providing a separator between a first electrode and a second electrode in a continuous zigzag type; and (b) sealing opposite side surfaces of the separator with the first electrode or the second electrode of the step (a) interposed therebetween.
(16) The step (b) may include sealing opposite side surfaces of the separator with the first electrode interposed therebetween by using a first sealer and a second sealer.
(17) The step (b) may include sealing opposite side surfaces of the separator with the second electrode interposed therebetween by using a first sealer and a second sealer.
(18) The step (b) may include sealing each edge of the separator of an uppermost side and a lowermost side on which the first electrode, the second electrode and the separator are stacked.
(19) In the step (b), the separator may be sealed by using at least one of thermal welding, ultrasonic wave welding, laser bonding, and an adhesive.
(20) According to the exemplary embodiment of the prevent invention, the separator may be sealed at the upper side of the negative electrode or the positive electrode in the process of stacking the negative electrode and the positive electrode while interposing the separator therebetween, to prevent a positive change of the negative electrode or the positive electrode, thereby improving stability.

Description

ADVANTAGEOUS EFFECTS
Description of the Drawings
(1) FIG. **1** illustrates a schematic perspective view of a rechargeable battery according to an exemplary embodiment of the present invention.
(2) FIG. **2** illustrates a schematic perspective view of the electrode assembly of FIG. **1**.
(3) FIG. **3** schematically illustrates main parts of a separator, a first electrode, and a second electrode which are disposed in a state where a portion of the electrode assembly of FIG. **2** is unfolded.

(4) FIG. **4** illustrates a top plan view schematically showing that a sealing portion is formed at an upper portion and a lower portion of a negative electrode in a state where a portion of an electrode assembly according to a first exemplary embodiment of the present invention is unfolded.
(5) FIG. **5** illustrates a side view schematically showing main parts of the electrode assembly according to the first exemplary embodiment of the present invention in which the sealing portion is formed.
(6) FIG. **6** illustrates a top plan view schematically showing that a sealing portion is formed at an upper portion and a lower portion of a positive electrode in a state where a portion of the electrode assembly according to a second exemplary embodiment of the present invention is unfolded.
(7) FIG. **7** illustrates a side view schematically showing main parts of the electrode assembly according to the second exemplary embodiment of the present invention in which the sealing portion is formed.
(8) FIG. **8** illustrates a top plan view schematically showing that a sealing portion is formed in a state where a portion of the electrode assembly according to a third exemplary embodiment of the present invention is unfolded.
(9) FIG. **9** illustrates a side view schematically showing main parts of the electrode assembly according to the third exemplary embodiment of the present invention in which the sealing portion is formed.
(10) FIG. **10** illustrates a flowchart schematically showing an electrode assembly manufacturing method according to the first exemplary embodiment of the present invention.
(11) FIG. **11** illustrates a side view schematically showing a state in which a first sealer and a second sealer of FIG. **10** are installed.
(12) FIG. **12** illustrates a flowchart schematically showing an electrode assembly manufacturing method according to the second exemplary embodiment of the present invention.
MODE FOR INVENTION
(13) The present invention will be described more fully hereinafter with reference to the accompanying drawings, in which exemplary embodiments of the invention are shown. As those skilled in the art would realize, the described embodiments may be modified in various different ways, all without departing from the spirit or scope of the present invention.
(14) The drawings and description are to be regarded as illustrative in nature and not restrictive. Like reference numerals designate like elements throughout the specification.
(15) Throughout this specification and the claims that follow, when it is described that an element is “coupled/connected” to another element, the element may be “directly coupled/connected” to the other element or “indirectly coupled/connected” to the other element through a third element. In addition, unless explicitly described to the contrary, the word “comprise” and variations such as “comprises” or “comprising” will be understood to imply the inclusion of stated elements but not the exclusion of any other elements.
(16) Throughout this specification and the claims that follow, it will be understood that when an element such as a layer, film, region, or substrate is referred to as being “on” another element, it can be directly on the other element or intervening elements may also be present. Further, in the specification, the word “˜on” or “˜over” means positioning on or below the object portion, and does not essentially mean positioning on the upper side of the object portion based on a gravity direction.
(17) FIG. **1** illustrates a schematic perspective view of a rechargeable battery according to an exemplary embodiment of the present invention, FIG. **2** illustrates a schematic perspective view of the electrode assembly of FIG. **1**, and FIG. **3** schematically illustrates main parts of a separator, a first electrode, and a second electrode which are disposed in a state where a portion of the electrode assembly of FIG. **2** is unfolded.
(18) As illustrated in FIG. **1** to FIG. **3**, according to a first exemplary embodiment of the present invention, a rechargeable battery **100** includes an electrode assembly **10** configured to including a

separator **15**, a first electrode **11**, and a second electrode **13**; a case **40** configured to accommodate the electrode assembly **10** therein; lead tabs **21** and **23** configured to include a first current collecting tab **21** connected with the first electrode **11**, and a second current collecting tab **23** connected with the second electrode **13**; a first electrode lead **31** connected with the first current collecting tab **21**; and a second electrode lead **33** connected with the second current collecting tab **23**. The first current collection tab **21** and the second current collection tab **23** may protrude in the same direction. Alternatively, the first current collection tab **21** and the second current collection tab **23** may protrude in opposite directions.
(19) For example, in the electrode assembly **10**, the first electrode (hereinafter referred to as a “negative electrode”) **11** and the second electrode (hereinafter referred to as a “positive electrode”) **13** may be disposed on opposite surfaces of the separator **15**, and the negative electrode **11**, the separator **15**, and the positive electrode **13** may be folded a predetermined number of times in a zigzag form.
(20) In the present exemplary embodiment, the electrode assembly **10** may be formed to have a zigzag shape by stacking the negative electrode **11** and the positive electrode **13** which are respectively inserted into spaces formed by folding the separator **15** the predetermined number of times.
(21) In other words, the negative electrode **11** and the positive electrode **13**, which are disposed at opposite sides of the separator **15**, may be respectively inserted into first insertion portions **11***a* and second insertion portions **13***a* formed in the separator **15** that is bent in the zigzag shape.
(22) The negative electrode **11** is disposed on one side of the separator **15**, and the first current collecting tab **21** (hereinafter referred to as “negative electrode current collecting tab”) is connected to an uncoated region thereof. The negative electrode current collecting tab **21** is connected to each negative electrode **13**, and is installed to connect the negative electrode **11** to the first electrode lead **31**.
(23) This negative electrode **11** may be disposed inside of the first insertion portions **11***a* formed in the spaces between portions of the separator **15** that are bent in the zigzag form. The negative electrode **11** may be formed by a stamping operation process by a press or the like, and a plurality of negative electrodes **11** may be disposed on a side surface of the separator **15**.
(24) The positive electrode **13** may be disposed on the other side of the separator **15** at a position opposite to the negative electrode **11** with the separator **15** therebetween.
(25) The positive electrode **13** may be disposed on the side surface of the separator **15** in a rectangular shape by a punching operation process. Although the positive electrode is described to be formed to have a rectangular shape in the present exemplary embodiment as an example, the present invention is not limited thereto, and the positive electrode **13** may be formed in various shapes such as a round shape in a portion thereof. In an implementation, at least one of the positive electrode and the negative electrode may be formed to have a polygon shape (an atypical polygon shape, e. g., ‘L’ shape, ‘U’ shape, or ‘T’ shape).
(26) The positive electrode **13** may be disposed inside of the second insertion portions **13***a* formed in the spaces between portions of the separator **15** that are bent in the zigzag form. The negative electrode **11** may be formed by a stamping operation process by a press or the like, and a plurality of negative electrodes **11** may be disposed on a side surface of the separator **15**.
(27) The positive electrode **13** may include a plurality of positive electrodes **13**, which are disposed on the side surface of the separator **15** in a state of being spaced apart from each other, and the second current collecting tab **23** may be connected thereto at each edge in a protruding manner.
(28) As such, the negative electrode **11** and the positive electrode **13**, which are disposed at opposite sides of the separator **15**, may be respectively inserted into the first insertion portions **11***a* and the second insertion portions **13***a* formed in the separator **15** that is bent in the zigzag shape.
(29) The first inserting portion **11***a* indicates a portion formed on one side surface of the separator **15** that is bent in the zigzag form, into which the negative electrode **11** is to be inserted. Similarly,

the second insertion portion **13***a* indicates a portion formed on the other surface of the separator **15** that is bent in the zigzag form, into which the positive electrode **13** is to be inserted.
(30) Meanwhile, the opposite surfaces of the separator **15** may be bonded to each other by a sealing portion **20** in a process of inserting the negative electrode **11** and the positive electrode **13** into the first insertion portion **11***a* and the second insertion portion **13***a*, respectively. This will be described in detail below.
(31) FIG. **4** illustrates a top plan view schematically showing that a sealing portion is formed at an upper portion and a lower portion of a negative electrode in a state where a portion of an electrode assembly according to a first exemplary embodiment of the present invention is unfolded, and FIG. **5** illustrates a side view schematically showing main parts of the electrode assembly according to the first exemplary embodiment of the present invention in which the sealing portion is formed.
(32) As illustrated in FIG. **4** and FIG. **5**, the negative electrode **11** may be stacked in a state where the negative electrode **11** is inserted into the zigzag bent surface of the first inserting portion **11***a*. Herein, the opposite side surfaces of the separator **15** at an upper side and a lower side of the negative electrode **11** with the negative electrode **11** therebetween may be bonded to each other by the sealing portion **20**.
(33) The sealing portion **20** bonds the upper side and the lower side of the negative electrode **11** at an open side of the first insertion portion **11***a* in the separator **15**, and thus the negative electrode **11** may be restrained in a state in which a position thereof is fixed inside of the first insertion portion **11***a*.
(34) Accordingly, the negative electrode **11** is stably positioned in the first inserting portion **11***a* by the sealing portion **20** even when vibration is generated by an impact from the outside of the rechargeable battery **100**, and thus it is possible to effectively prevent a short circuit caused by a change in a position of the negative electrode **11**.
(35) The sealing portion **20** is exemplarily illustrated as being bonded to the opposite side surfaces of the separator **15** by thermal welding in the present exemplary embodiment.
(36) However, the bonding of the sealing portion **20** is not necessarily limited to thermal welding, and may be applied to any one of ultrasonic wave welding, laser bonding, and an adhesive.
(37) As described above, in the electrode assembly **10** of the secondary battery **100** of the present exemplary embodiment, the negative electrode **11** and the positive electrode **13** are stacked with the separator **15** interposed therebetween, while the separator **15** is provided in a zigzag staggered manner.
(38) Herein, since the opposite surfaces of the separators **15** with the negative electrode **11** or the positive electrode **13** therebetween are bonded to each other, the negative electrodes **11** and the positive electrodes may be stably positioned inside of the separator **15**. Therefore, the negative electrode **11** and the positive electrode **13** may be prevented from being moved while inserting the electrode assembly **10** into the case **40**. Also, the negative electrode **11** and positive electrode **13** may be prevented from being shifted to suppress a short circuit, thereby improving stability of the rechargeable battery **100**.
(39) FIG. **6** illustrates a top plan view schematically showing that a sealing portion is formed at an upper portion and a lower portion of a positive electrode in a state where a portion of the electrode assembly according to a second exemplary embodiment of the present invention is unfolded, and FIG. **7** illustrates a side view schematically showing main parts of the electrode assembly according to the second exemplary embodiment of the present invention in which the sealing portion is formed. The same reference numerals as those of FIG. **1** to FIG. **5** denote the same or similar members having the same or similar functions. Hereinafter, detailed descriptions of the same reference numerals will be omitted.
(40) As illustrated in FIG. **6** and FIG. **7**, according to the second exemplary embodiment of the prevent invention, a sealing portion **120** of an electrode assembly **110** seals an upper portion and a lower portion of the positive electrode **13** inserted into the second insertion portion **13***a* on the

zigzag-bent surface of the separator **15**.
(41) Accordingly, the positive electrode **13** is stably positioned in the second insertion portion **13***a* by the sealing portion **120** even when vibration is generated by an impact from the outside of the rechargeable battery **100**, and thus it is possible to effectively prevent a short circuit caused by a change in a position of the positive electrode **13**.
(42) FIG. **8** illustrates a top plan view schematically showing that a sealing portion is formed in a state where a portion of the electrode assembly according to a third exemplary embodiment of the present invention is unfolded, and FIG. **9** illustrates a side view schematically showing main parts of the electrode assembly according to the third exemplary embodiment of the present invention in which the sealing portion is formed. The same reference numerals as those of FIG. **1** to FIG. **7** denote the same or similar members having the same or similar functions. Hereinafter, detailed descriptions of the same reference numerals will be omitted.
(43) As illustrated in FIG. **8** and FIG. **9**, according to the third exemplary embodiment of the present invention, in a sealing portion **220** of an electrode assembly **210** an upper edge portion and a lower edge portion of the zigzag-bent separator **15** are bonded in a state where the negative electrode **11** and the positive electrode **13** are respectively disposed inside of the first insertion portion **11***a* and the second insertion portion **13***a*.
(44) As a result, according to the present exemplary embodiment, the electrode assembly **210** may effectively prevent a short circuit caused by changes in positions where the negative electrode **11** and the positive electrode **13** are stacked when a vibration is transferred thereto by external impact, by connecting the upper edge portion and the lower edge portion of the separator **15** in a state where the separator **15**, the negative electrode **11**, the positive electrode **13** are stacked as unit bodies.
(45) FIG. **10** illustrates a flowchart schematically showing an electrode assembly manufacturing method according to the first exemplary embodiment of the present invention, and FIG. **11** illustrates a side view schematically showing a state in which a first sealer and a second sealer of FIG. **10** are installed. The same reference numerals as those of FIG. **1** to FIG. **9** denote the same or similar members having the same or similar functions. Hereinafter, detailed descriptions of the same reference numerals will be omitted. Hereinafter, a rechargeable battery manufacturing method will be described in detail below.
(46) First, the first electrode **11** and the second electrode **13** are provided with the separator **15** interposed therebetween (S**10**). Hereinafter, the first electrode **11** serves as a negative electrode, and the second electrode **13** serves as a positive electrode.
(47) The separator **15** in step S**10** may be provided in a continuous zigzag fashion between the first electrode **11** and the second electrode **13**.
(48) Herein, the first electrode **11** may be provided to the first inserting portion **11***a* formed on one side of the zigzag bent portion of the separator **15**, and the second electrode **13** may be provided to the second insertion portion **13***a* formed on the other side of the zigzag bent portion of the separator **15**.
(49) Next, side surfaces of the separator **15** are sealed at an edge portion of the separator **15** with the first electrode **11** or the second electrode **13** interposed therebetween. For this purpose, a first sealer **111** and a second sealer **113** may be provided with the first electrode **11** or the second electrode **13** interposed therebetween, to bond the separator **15** with the upper and lower portions of the first electrode **11** or the second electrode **13**.
(50) In step S**20**, the side surfaces of the separator **15** may be sealed at the upper portion and the lower portion of the first electrode **11** or the second electrode **13** in a state where the first electrode **11** and the second electrode **13** are respectively disposed inside of the first electrode **11** and the positive electrode **13** formed on one side and the other side of the separator **15** bent in a zigzag shape.
(51) In the present exemplary embodiment, step S**20** is exemplarily illustrated as sealing the

separator **15** at the upper portion and the lower portion of the first electrode **11** by the sealing portion **20** formed by using the first sealer **111** and the second sealer **113**. However, the sealing portion **20** is not necessarily limited to sealing the upper portion and the lower portion of the first electrode **11**, and the upper portion and the lower portion of the second electrode **13** may be sealed by using the first sealer **111** and the second sealer **113**. Thus, in step S**20**, at least one of the first electrode **11** and the second electrode **13** may be restrained in a state in which a position thereof is fixed inside of the separator **15**.
(52) Herein, in step S**20**, the sealing of the separator **15** may be performed by using thermal welding, ultrasonic wave welding, laser bonding, an adhesive, or an adhesive tape.
(53) After step S**20**, a finishing tape at least partially may surround the electrode assembly **10**. Next, the electrode assembly **10** may be accommodated into the case **40**, as illustrated in FIG. **1**.
(54) FIG. **12** illustrates a flowchart schematically showing an electrode assembly manufacturing method according to the second exemplary embodiment of the present invention. The same reference numerals as those of FIG. **1** to FIG. **11** denote the same or similar members having the same or similar functions. Hereinafter, detailed descriptions of the same reference numerals will be omitted. Hereinafter, a rechargeable battery manufacturing method will be described in detail below.
(55) First, the first electrode **11** and the second electrode **13** are provided with the separator **15** interposed therebetween (S**110**). The separator **15** in step S**110** may be provided in a continuous zigzag fashion between the first electrode **11** and the second electrode **13**.
(56) Next, an uppermost edge portion and the lowermost edge portion of the separator **15** that is bent in the zigzag shape are sealed with each other in the electrode assembly **10** in which the first electrode **11** and the second electrode **13** are stacked as a unit body with the separator **15** interposed therebetween (S**120**).
(57) That is, the sealing may be performed in a state where the first electrode **11** and the second electrode **13** are disposed inside of the separator **15**, by respectively disposing the first sealer **111** and the second sealer **113** at an uppermost end and a lowermost side of the electrode assembly **10**.
(58) As such, since the sealing is performed between the uppermost end and the lowermost end of the separator **15** in a state where electrode assemblies **10** are stacked as a unit body, the first electrode **11** and the second electrode **13** may be stably stacked so as to prevent short circuit occurrence.
(59) While this invention has been described in connection with what is presently considered to be practical exemplary embodiments, it is to be understood that the invention is not limited to the disclosed embodiments, but, on the contrary, is intended to cover various modifications and equivalent arrangements included within the spirit and scope of the appended claims.
DESCRIPTION OF SYMBOLS
(60) **10** . . . assembly **11** . . . negative electrode **11***a* . . . first insertion portion **13** . . . positive electrode **13***a* . . . second insertion portion **15** . . . separator **20** . . . portion **21** . . . first current collecting tab **23** . . . second current collecting tab **31** . . . first electrode Lead **33** . . . second electrode lead **111** . . . first sealer **113** . . . second sealer

Claims

1. A rechargeable battery comprising: an electrode assembly; and a case configured to accommodate the electrode assembly, wherein the electrode assembly comprises: a separator configured to include a first insertion portion formed in a first direction and a second insertion portion formed in a second direction, which are alternately stacked; a first electrode inserted into the first insertion portion; a second electrode inserted into the second insertion portion while the separator is disposed between the first electrode and the second electrode; a sealing portion configured to bond opposite surfaces of the separator into which the first electrode and the second

electrode are inserted with the separator interposed therebetween; and a lead tab configured to include a first current collecting tab connected to the first electrode and a second current collecting tab connected to the second electrode, wherein the first current collecting tab includes a first portion covered with the separator and a second portion drawn out of the separator in a third direction different from the first direction, and wherein the sealing portion is disposed on a first side of the separator from which the first current collecting tab is drawn out, and is further disposed on a second side of the separator opposite from the first side and opposite of the third direction.

2. The rechargeable battery of claim 1, wherein the separator is formed as a continuous zigzag type between the first electrode and the second electrode, the separator in which the first insertion portion and the second insertion portion are alternatively formed.

3. The rechargeable battery of claim 2, wherein the sealing portion bonds opposite surfaces of the separator with the first electrode interposed therebetween at the first insertion portion.

4. The rechargeable battery of claim 2, wherein the sealing portion bonds opposite surfaces of the separator with the second electrode interposed therebetween at the second insertion portion.

5. The rechargeable battery of claim 2, wherein the sealing portion bonds edges of the separator in which the first electrode and the second electrode are stacked in plural.

6. The rechargeable battery of claim 5, wherein the sealing portion bonds each edge of the separator of an uppermost side and a lowermost side on which the first electrode, the second electrode and the separator are stacked.

7. The rechargeable battery of claim 1, wherein the sealing portions bond the separator by using at least one of thermal welding, ultrasonic wave welding, laser bonding, an adhesive, and an adhesive tape.

8. The rechargeable battery of claim 1, wherein at least one of the first electrode and the second electrode has a polygon shape.

9. The rechargeable battery of claim 1, further comprising: a first electrode lead connected to the first current collecting tab; and a second electrode lead connected to the second current collecting tab.

10. A rechargeable battery manufacturing method comprising: providing a separator between a first electrode and a second electrode in a continuous zigzag type, wherein the separator includes a first insertion portion formed in a first direction and receiving the first electrode, and a second insertion portion formed in a second direction and receiving the second electrode; and sealing opposite side surfaces of the separator with the first electrode or the second electrode interposed therebetween, wherein the sealing includes sealing a first portion of a first side of the separator on which a current collecting tab connected to the first electrode is drawn out, wherein the current collecting tab is drawn out in a third direction different from the first direction, and sealing a second portion of a second side of the separator opposite from the first side and opposite the third direction.

11. The rechargeable battery manufacturing method of claim 10, wherein the opposite side surfaces of the separator are sealed with the first electrode interposed therebetween by using a first sealer and a second sealer.

12. The rechargeable battery manufacturing method of claim 10, wherein the opposite side surfaces of the separator are sealed with the second electrode interposed therebetween by using a first sealer and a second sealer.

13. The rechargeable battery manufacturing method of claim 10, wherein the sealing includes sealing each edge of the separator of an uppermost side and a lowermost side on which the first electrode, the second electrode, and the separator are stacked.

14. The rechargeable battery manufacturing method of claim 10, wherein the separator is sealed by using at least one of thermal welding, ultrasonic wave welding, laser bonding, an adhesive, and an adhesive tape.

15. The rechargeable battery manufacturing method of claim 10, further comprising:

accommodating an electrode assembly in a case, the electrode assembly including the first electrode, the second electrode, and the separator.